Province of the
EASTERN CAPE
EDUCATION

# NATIONAL SENIOR CERTIFICATE

## GRADE 10

## NOVEMBER 2018

# TECHNICAL MATHEMATICS P2

**MARKS:** **100**

**TIME:** **2 hours**

This question paper consists of 10 pages and a special answer book.

## INSTRUCTIONS AND INFORMATION

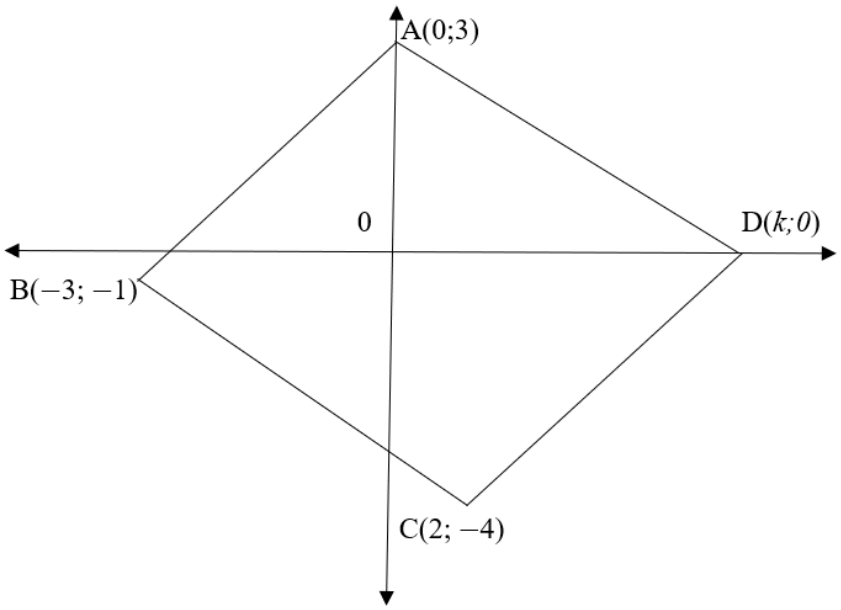
Read the following instructions carefully before answering the questions.

1. This question paper consists of 8 questions.
2. Answer ALL the questions in the SPECIAL ANSWER BOOK provided.
3. Clearly show ALL calculations, diagrams, graphs, et cetera which you have used in determining the answers.
4. Answers only will NOT necessarily be awarded full marks.
5. You may use an approved scientific calculator (non-programmable and non-graphical) unless stated otherwise.
6. If necessary, round off your answers to TWO decimal places, unless stated otherwise.
7. Diagrams are not necessarily drawn to scale.
8. Write neatly and legibly.

Copyright reserved 

## QUESTION 1

1.1 In the diagram below A (0; 3), B (−3; −1); C (2; −4) and D (*k*; 0) are the vertices of the quadrilateral ABCD.




1.1.1 Determine the coordinates of M, the midpoint of AC. (2)

1.1.2 Determine the gradient of MB. (2)

1.1.3 Hence, calculate the equation of the straight line passing through M and B. (3)

1.1.4 If the length of CD = 5, show that k = 5. (4)

1.1.5 If AB = CD, what type of a quadrilateral is ABCD? Justify your answer by relevant calculations. (4)
**[15]**

## QUESTION 2

2.1 If $\hat{A} = 57^{o}$ and $\hat{B} = 39^{o}$, use a calculator to evaluate the following:

2.1.1 $cosec\ \hat{A} + cot\ \hat{B}$ (3)

2.1.2 $2cos\left(\frac{3\hat{A}}{2}\right)$ (2)

2.2 Given $5cos\theta = -3$ and $180^{0} \leq \theta \leq 270^{0}$, determine the value of the following **with the aid of a sketch:**

2.2.1 $cos\,\theta + tan\,\theta$ (4)

2.2.2 $sec\,\theta$ (3)

2.3 Solve for $x$, if $2tan\,(2x + 12°) - 3 = 1,\ x \in [0°; 90°]$. (5)

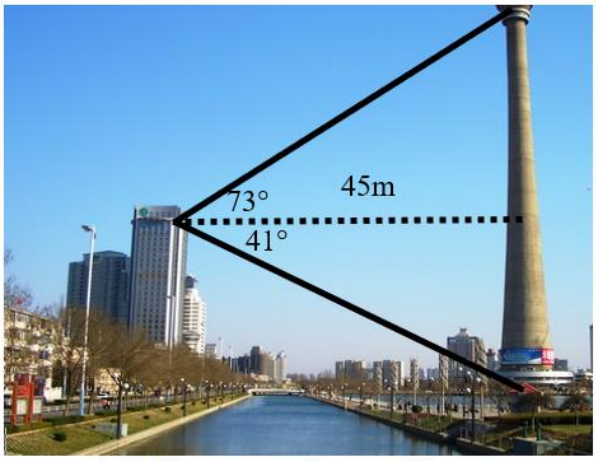
**[17]**

## QUESTION 3

Study the picture and the diagram below.

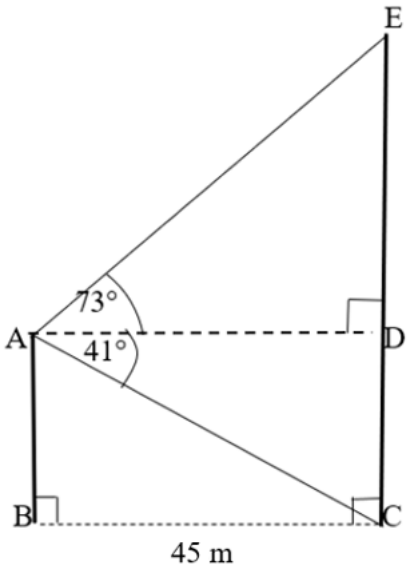
- The foot of the tower EC and the bottom of the building AB lie on the same horizontal plane.
- The angle of elevation from the top of the building to the top of the tower is 73° and the angle of depression to the foot of the tower is 41°.
- The distance between the tower and the building is 45 m.

Answer the following questions, correct to the nearest cm.




http://www.wonders-world.com

The 2D diagram below models the building and the tower:




3.1 Give, with reasons, the size of angle $A\hat{B}C$. (2)

3.2 Hence, determine AB, the height of the building. (4)

3.3 Calculate ED. How much higher is the tower than the building? (4)

3.4 Determine the height of the tower. (2)

**[12]**

## QUESTION 4

4.1 Consider the function:

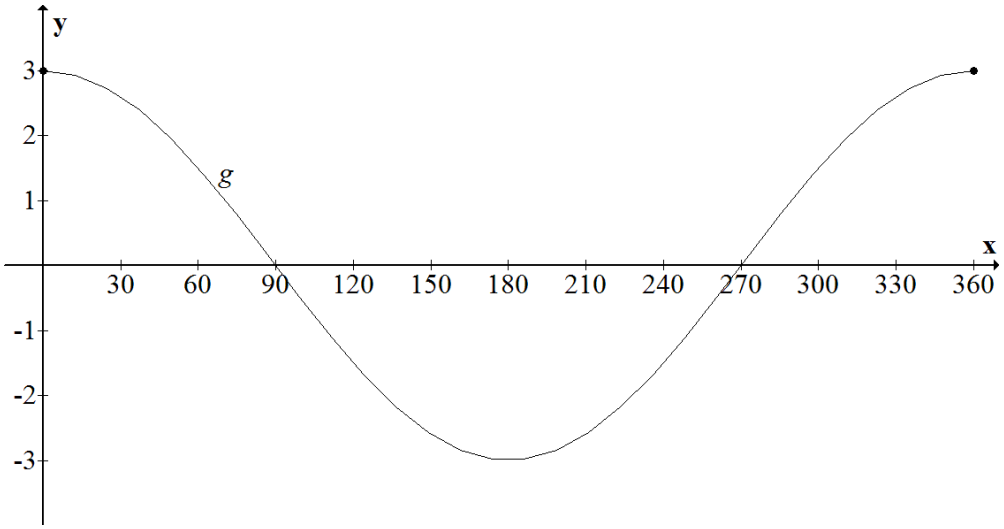
$$f(x) = 3tanx$$

4.1.1 Draw a neat sketch of $f(x) = 3tanx$ for $0^0 \leq x \leq 360^0$ on the axes provided in the SPECIAL ANSWER BOOK.
Clearly indicate, in your sketch, the intercepts with the axes and the asymptotes. (4)

4.1.2 Write down the period of $f(x) = 3tanx$. (2)

4.1.3 If the graph of $f(x) = 3tanx$ is reflected about the $x-axis$, write down the equation of the new graph, g($x$), obtained by this reflection. (1)

4.2 The diagram below shows the graph of $g(x) = acosx$ for $0^0 \leq x \leq 360^0$.




4.2.1 Determine the value of $a$. (2)

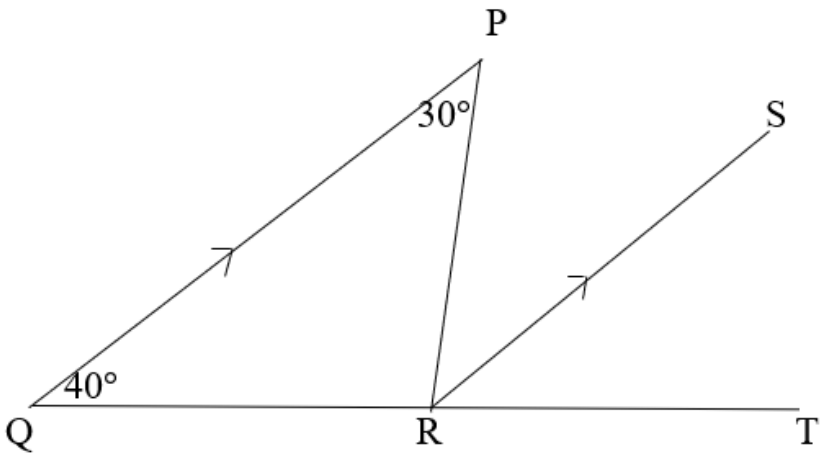
4.2.2 If the graph of $g$ is translated 2 units upwards to obtain a new graph $h$, write down the range of $h$. (2)

**[11]**

Copyright reserved 

## QUESTION 5

5.1 In the figure below, PQ//RS. $\hat{Q} = 40°$and $\hat{P} = 30°$.




Determine the following:

5.1.1 $P\hat{R}S$ (2)

5.1.2 $T\hat{R}S$ (2)

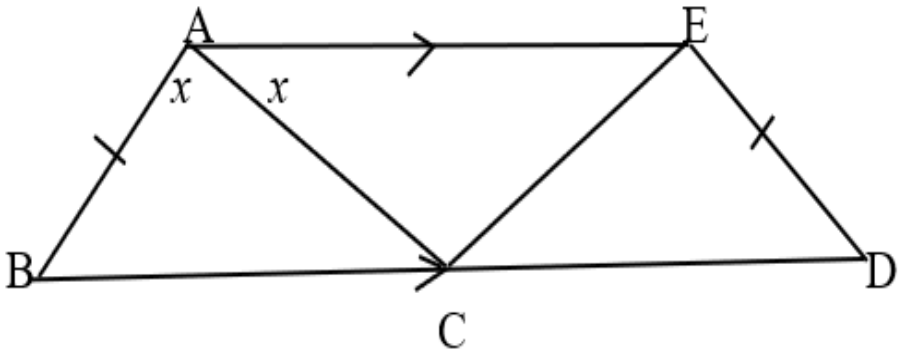
5.1.3 $P\hat{R}Q$ (2)

5.1.4 $P\hat{R}T$ (2)

5.1.5 What is the relationship between $\hat{P}, \hat{Q}$ and $P\hat{R}T$? (1)
**[9]**

**QUESTION 6**

Consider quadrilateral ABDE below. BD//AE and BD = 2AE. AB = ED and AC bisects $B\hat{A}E$. $B\hat{A}C = C\hat{A}E = x$.




6.1 What type of quadrilateral is ABDE? Justify your answer. (3)

6.2 Give, with reasons, the size of each of the angles listed below, in terms of $x$:

6.2.1 $\hat{D}$ (2)

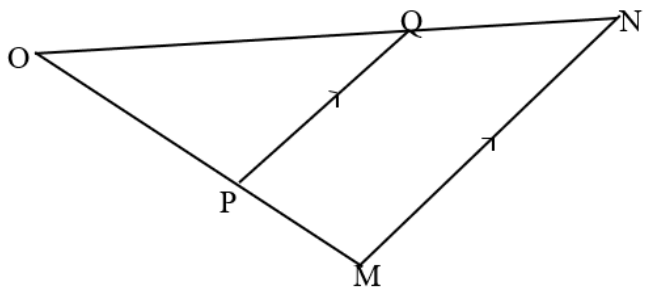
6.2.2 $A\hat{E}D$ (2)

6.3 For which value(s) of $x$ will ACDE be a parallelogram? Justify your answer. (2)

6.4 If $x = 60°$:
Classify quadrilateral ABCE and justify your answer. (4)
**[13]**

## QUESTION 7

7.1 In the diagram below, line MN is parallel to line PQ.




7.1.1 Prove that $\Delta PQO$ and $\Delta MNO$ are similar. (3)

7.1.2 Hence, give the mathematical relationship of the sides, based on the proof from QUESTION 7.1.1, above. (1)

7.2 It is also given that $MN = 9$ units and $PQ = 6$ units.

7.2.1 If $ON = 12$ units, how long is OQ? Show ALL calculations. (3)

7.2.2 If $OP = 19$ units, how long is PM? Show ALL calculations. (3)

**[10]**

## QUESTION 8

8.1 Convert the following:

8.1.1 107,5° to degrees-minute-second (3)

8.1.2 69°64'89" to degrees (3)

8.2 What is the measure in degrees of central angle $\theta$ that intersect an arc length of 35 cm on a circle with the radius of 7 cm? (4)

8.3 Simplify, without using a calculator. Answer must be in degrees.

$$2\pi - \frac{\pi}{9} - 120°$$

(3)
**[13]**

**TOTAL: 100**

Copyright reserved

Province of the
EASTERN CAPE
EDUCATION

| | |
|---|---|
| **LEARNER'S SURNAME/** ***NAAM VAN LEERDER:*** | |

| | |
|---|---|
| **LEARNER'S SURNAME/** ***NAAM VAN LEERDER:*** | |

# NATIONAL SENIOR CERTIFICATE/ *NASIONALE SENIOR SERTIFIKAAT*

## GRADE/*GRAAD* 10

## NOVEMBER 2018

## TECHNICAL MATHEMATICS P2/*TEGNIESE WISKUNDE V2*
## SPECIAL ANSWER BOOK/*SPESIALE ANTWOORDEBOEK*

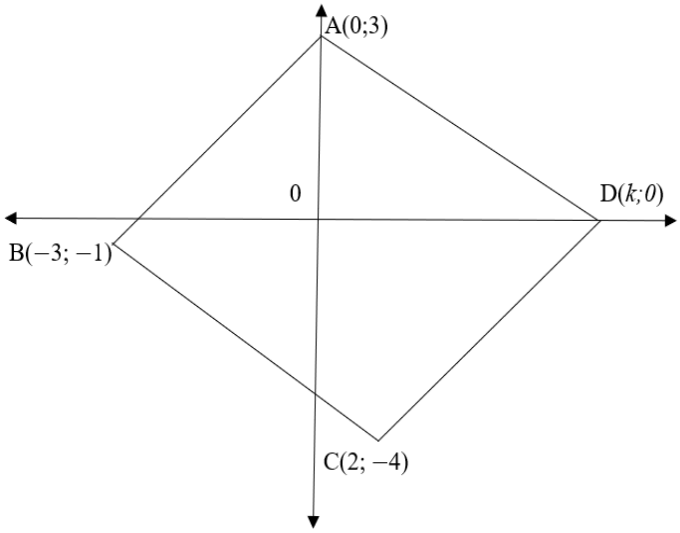
| QUESTION/*VRAAG* | MARKS/*PUNTE* | | | SCHOOL/*SKOOL* (Level/*Vlak 1 mod.*) | | | DISTRICT/*DISTRIK* (Level/*Vlak 2 mod.*) | | | PROVINCIAL/*PROVINSIAAL* (Level/*Vlak 3 mod.*) | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **1** | | | | | | | | | | | | |
| **2** | | | | | | | | | | | | |
| **3** | | | | | | | | | | | | |
| **4** | | | | | | | | | | | | |
| **5** | | | | | | | | | | | | |
| **6** | | | | | | | | | | | | |
| **7** | | | | | | | | | | | | |
| **8** | | | | | | | | | | | | |
| **TOTAL/*TOTAAL*** | | | | | | | | | | | | |

This answer book consists of 12 pages./*Hierdie antwoordeboek bestaan uit 12 bladsye.*

| FOLLOW THESE INSTRUCTIONS CAREFULLY | VOLG HIERDIE INSTRUKSIES NOUKEURIG |
|---|---|
| 1. Answer ALL questions in the spaces provided. | 1. Beantwoord ALLE vrae in die ruimtes wat voorsien is. |
| 2. No pages may be torn from this answer book. | 2. Geen bladsye mag uit hierdie antwoordeboek geskeur word nie. |
| 3. Answers must be written in black/blue ink as distinctly as possible. Do not write in the margins. | 3. Skryf die antwoorde so duidelik moontlik met swart/blou ink. Moenie in die kantlyn skryf nie. |
| 4. Indicate the questions you have answered by drawing a circle around the relevant numbers on the front cover of the answer book where marks are to be recorded. | 4. Dui die vrae wat jy beantwoord het aan op die voorblad van die antwoordeboek waar die punte aangebring is, deur 'n kringetjie te trek om die nommers van die vrae wat jy beantwoord het. |
| 5. Rule off through any work/rough work that must not be marked. | 5. Trek 'n netjiese lyn deur enige werk/rofwerk wat nie nagesien moet word nie. |
| 6. In the event that you use the additional space provided:<br>6.1 Write down the number of the question.<br>6.2 Leave a line and rule off after your answer. | 6. Ingeval jy die bykomende ruimte wat voorsien word, gebruik:<br>6.1 Skryf die nommer van die vraag neer.<br>6.2 Laat 'n lyn oop en trek 'n lyn na jou antwoord. |

Copyright reserved/*Kopiereg voorbehou* 

## QUESTION/*VRAAG* 1



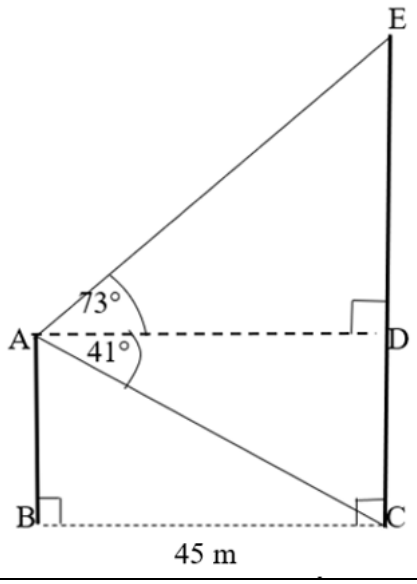

| | **Solution/*Oplossing*** | **Marks/ *Punte*** |
|---|---|---|
| 1.1.1 | | (2) |
| 1.1.2 | | (2) |
| 1.1.3 | | (3) |
| 1.1.4 | | (4) |
| 1.1.5 | | (4) |
| | | **[15]** |

## QUESTION/*VRAAG* 2

| | **Solution/*Oplossing*** | **Marks** ***Punte*** |
|---|---|---|
| 2.1.1 | | (3) |
| 2.1.2 | | (2) |
| 2.2.1 | | (4) |
| 2.2.2 | | (3) |
| 2.3 | | (5) |
| | | **[17]** |

Copyright reserved/*Kopiereg voorbehou*

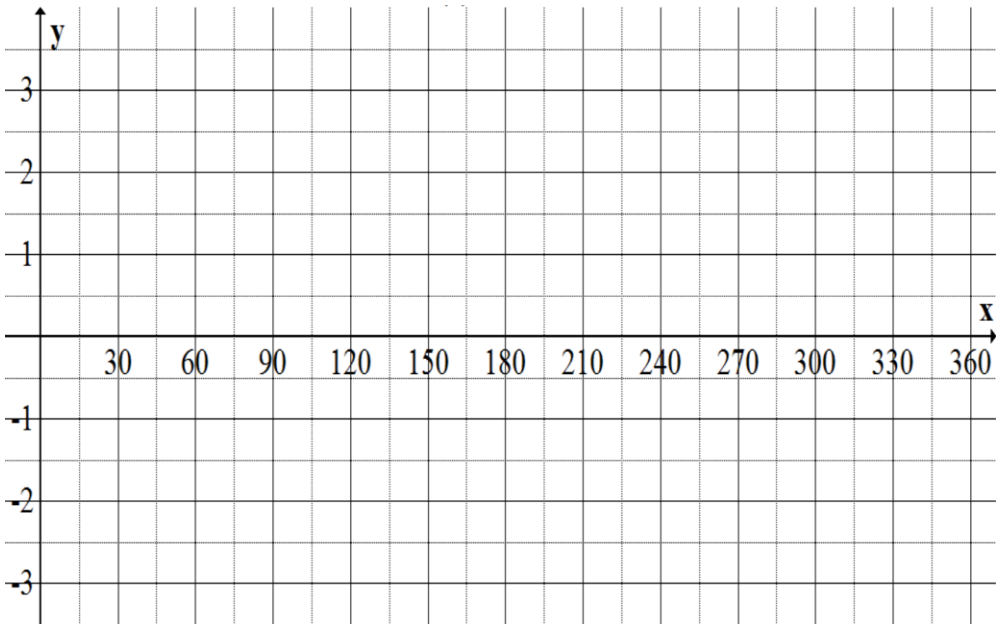
## QUESTION/*VRAAG* 3

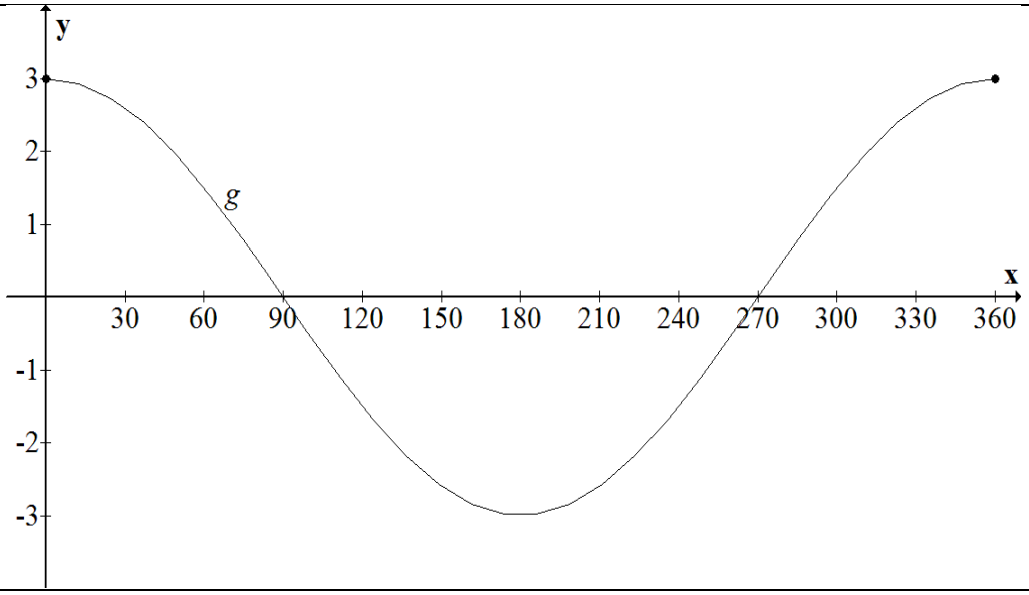



| | **Solution/*Oplossing*** | **Marks/ *Punte*** |
|---|---|---|
| 3.1 | | |
| | | (2) |
| 3.2 | | |
| | | |
| | | |
| | | |
| | | |
| | | |
| | | (4) |
| 3.3 | | |
| | | |
| | | |
| | | (4) |
| 3.4 | | |
| | | |
| | | (2) |
| | | **[12]** |

Copyright reserved/*Kopiereg voorbehou* 

## QUESTION/*VRAAG* 4

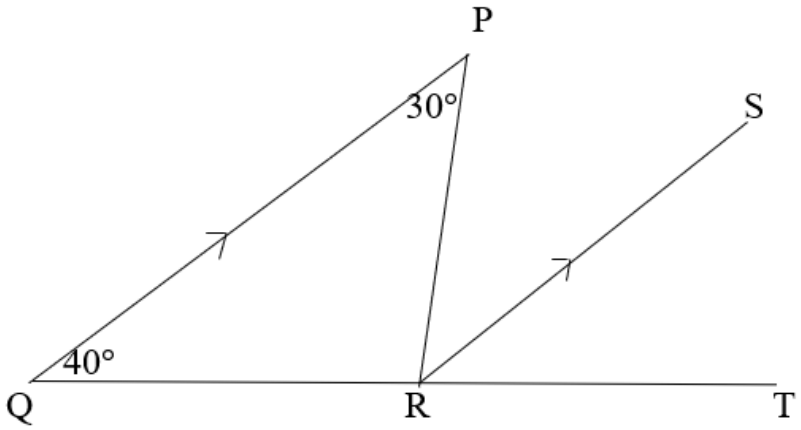
| | **Solution/*Oplossing*** | **Marks** ***Punte*** |
|---|---|---|
| 4.1.1 |  | (4) |
| 4.1.2 | | (2) |
| 4.1.3 | | (1) |




| | | |
|---|---|---|
| 4.2.1 | | (2) |
| 4.2.2 | | (2) |
| | | **[11]** |

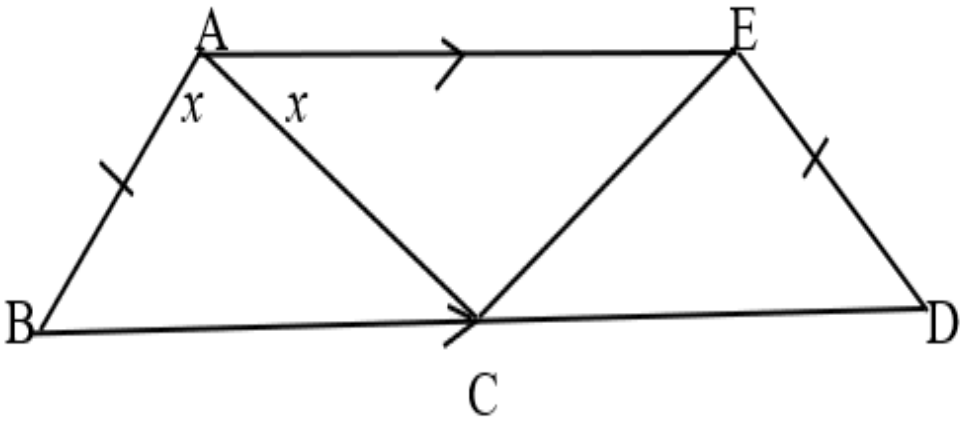
Copyright reserved/*Kopiereg voorbehou* 

## QUESTION/*VRAAG* 5




| | **Solution/*Oplossing*** | **Marks** ***Punte*** |
|---|---|---|
| 5.1.1 | | (2) |
| 5.1.2 | | (2) |
| 5.1.3 | | (2) |
| 5.1.4 | | (2) |
| 5.1.5 | | (1) |
| | | **[9]** |

**QUESTION/*VRAAG* 6**




| | **Solution/*Oplossing*** | **Marks/ *Punte*** |
|---|---|---|
| 6.1 | | (3) |
| 6.2.1 | | (2) |
| 6.2.2 | | (2) |
| 6.3 | | (2) |
| 6.4 | | (4) |
| | | **[13]** |

## QUESTION/*VRAAG* 7

| **Solution/*Oplossing*** | | **Marks** ***Punte*** |
|---|---|---|
| 7.1.1 | | (3) |
| 7.1.2 | | (1) |
| 7.2.1 | | (3) |
| 7.2.2 | | (3) |
| | | **[10]** |

Copyright reserved/*Kopiereg voorbehou*

## QUESTION/*VRAAG* 8

| | **Solution/*Oplossing*** | **Marks** ***Punte*** |
|---|---|---|
| 8.1.1 | | (3) |
| 8.1.2 | | (3) |
| 8.2 | | (4) |
| 8.3 | | (3) |
| | | **[13]** |

Copyright reserved/*Kopiereg voorbehou*

| | **Additional Space/*Addisionele Ruimte*** | **Marks** ***Punte*** |
|---|---|---|
| | | |

| | **Additional Space/*Addisionele Ruimte*** | **Marks** ***Punte*** |
|---|---|---|
| | | |
| | | |
| | | |
| | | |
| | | |
| | | |
| | | |
| | | |
| | | |
| | | |
| | | |
| | | |
| | | |
| | | |
| | | |
| | | |
| | | |
| | **TOTAL/*TOTAAL:*** | **100** |

Copyright reserved/*Kopiereg voorbehou*